

**CAT 3 Deadline 15<sup>th</sup> September 2025 at 11.59pm (50 marks)**

**Q1.** A clinic has one doctor, and patients arrive according to a Poisson process at a rate of **6 patients/hour**. The service time follows an exponential distribution with an average service rate of **8 patients/hour**.

Compute the

- a) Utilization
- b) Mean number in system
- c) Mean number in queue
- d) Mean waiting time in system
- e) Mean waiting time in queue
- f) Probability system empty
- g) Probability of more than 4 customers in the system

**Q2.** At a help desk, customers arrive at a rate of **3 per min**, and the clerk can serve them at an average rate of **12 per hour**.

Compute the

- a) Utilization
- b) Mean number in system
- c) Mean number in queue
- d) Mean waiting time in system
- e) Mean waiting time in queue
- f) Probability system empty
- g) Probability of more than 4 customers in the system

**Q3.** Server utilization-A single-lane toll booth receives cars at a rate of **1 per hour**, and the operator can process cars at **20 per min**.

Compute the

- a) Utilization
- b) Mean number in system
- c) Mean number in queue
- d) Mean waiting time in system
- e) Mean waiting time in queue
- f) Probability system empty
- g) Probability of more than 4 customers in the system

**Q4.** In a small post office, arrivals occur at a rate of **5 per hour**, and service rate is **6 per hour**.

Compute the

- a) Utilization
- b) Mean number in system
- c) Mean number in queue
- d) Mean waiting time in system
- e) Mean waiting time in queue
- f) Probability system empty
- g) Probability of more than 4 customers in the system

**Q5.** In a call center with one operator, calls arrive at a rate of **2 per minute**, and the operator serves at **3 per minute**.

Compute the

- a) Utilization
- b) Mean number in system
- c) Mean number in queue
- d) Mean waiting time in system
- e) Mean waiting time in queue
- f) Probability system empty
- g) Probability of more than 4 customers in the system

**Q6.** Using Chapman-Kolmogorov assumptions, derive Poisson process, hence compute the mean and the variance the resulting Poisson Process (Show all your working)